

What happens to the inhibitory control functions of the right inferior frontal cortex when this area is dominant for language?

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Abstract

A low number of individuals show an atypical brain control of language functions that differs from the typical lateralization in the left cerebral hemisphere. In these cases, the neural distribution of other cognitive functions is not fully understood. Although there is a bias towards a mirrored brain organization consistent with the Causal hypothesis, some individuals are found to be exceptions to this rule. However, no study has focused on what happens to the homologous language areas in the right frontal inferior cortex. Using an fMRI-adapted stop-signal task in a healthy non right-handed sample (50 typically lateralized and 36 atypically lateralized for language production), our results show that atypical lateralization is associated with a mirrored brain organization of the inhibitory control network in the left hemisphere: inferior frontal cortex, presupplementary motor area, and subthalamic nucleus. However, the individual analyses revealed a large number of cases with a noteworthy overlap in the inferior frontal gyrus, which shared both inhibitory and language functions. Further analyses showed that atypical lateralization was associated with stronger functional and structural interhemispheric connectivity.

Importantly, we did not find task performance differences as a function of lateralization, but there was an association between atypical dominance in the inferior frontal cortex and higher scores on schizotypy and autistic spectrum traits, as well as worse performance on a reading accuracy test. Together, these results partially support the Causal hypothesis of hemispheric specialization and provide further evidence of the link between atypical hemispheric lateralization and increased interhemispheric transfer through the corpus callosum.

eLife assessment

This paper describes **important** advances from the results of studies on atypical language lateralisation and how this might impact language behaviour, as well as its associations with reading skills and the prevalence of schizotypy and autistic traits. Despite most of the conclusions of this paper being supported by evidence, several aspects including the characteristics of participants tested, hypotheses/predictions, and data analysis, are **incomplete** and/or need some clarification. The work has implications for evolutionary hypotheses and our understanding of the effects of brain damage on distinct cognitive functions.

Introduction

One of the oldest findings in human neuroscience is that the brain control of language is lateralized to the left hemisphere^{1–3}. However, not all individuals present a typical organization of this function. Some left-handers (22–24%) show an atypical lateralization of language – right or ambilateral^{4–6}. Also, this atypical organization is observed in some neurodevelopmental disorders such as schizophrenia, dyslexia, or autistic spectrum^{7–9}.

How these atypical individuals organize the rest of their lateralized functions has been an extensively researched question. Historically, two different hypotheses have been proposed to explain the hemispheric specialization of the brain. The Causal hypothesis^{10, 11} argues that rapid activities requiring the implementation of sequences of cognitive processes are better performed from a single, unilateral control. As a result, these activities are innately programmed to be lateralized, and, according to this model, some functions will be performed better if they are controlled by different hemispheres. The Statistical hypothesis¹², on the other hand, postulates that each cognitive function lateralizes independently from the others, making the probability of finding an atypical hemispheric dominance merely a statistical phenomenon. The evidence supporting each theory is mixed. Some studies have obtained data favoring the Statistical hypothesis when analyzing language, visuospatial attention, and face processing^{13–15}. Most investigations, however, have demonstrated a mirrored organization of praxis^{16, 17}, visuospatial attention^{14, 17, 18}, face recognition^{14, 17, 19}, and emotional prosody¹⁷ in most individuals with atypical lateralization of language. Nonetheless, it should be noted that this reversed pattern is not observed in all atypical participants, and all the studies found specific cases in favor of both theories¹⁷.

All these studies have employed a production task to investigate language lateralization in the inferior frontal cortex (IFC), classifying left-handed participants into typical and atypical groups. The rest of the lateralized cognitive functions have been examined using several tasks, but they all mainly depended on posterior parts of the brain. This may have limited the results, given that they were not dealing with directly homotopic functions, which would have been the most sensitive scenario to the proposed advantages of hemispheric specialization^{20, 21}. To directly address the language production function, which is typically dependent on the left IFC, we would have to use a cognitive task that is typically dependent on the homotopic right IFC. According to previous literature, this function would be inhibitory control. Extensive results support the notion that stopping an ongoing response involves a rightward network that includes the IFC, presupplementary motor area (preSMA), and subthalamic nucleus (STN)^{22–25}. Thus, inhibitory control and language production together are the ideal candidates to investigate the Causal vs. Statistical disjunctive. But surprisingly, no investigation to date has analyzed this.

In this study, we explored the relationship between the lateralization of language production and inhibitory control. To do so, we divided non-right-handed participants into typically lateralized and atypically lateralized for language using the verb generation task. These two groups then completed the stop-signal task. We hypothesized that the atypical group would show a mirrored brain organization during the stop-signal task. Additionally, we investigated if a relation existed between the lateralization of both functions and inter-hemispheric structural-functional connectivity²⁶, as well as with behavioral markers of certain clinical conditions that have been related with atypical lateralization^{7–9}.

Method

1. Participants

Eighty-six participants were included in the present study. They were selected following a functional magnetic resonance imaging (fMRI) language lateralization assessment via a verb generation task. Hence, 50 were typically lateralized for language – left-dominant – (mean $\pm$ SD age = 22.4 ± 3.6 years; 24 male, 26 female) and 36 were atypically lateralized (mean $\pm$ SD age = 23.4 ± 4 years; 16 male, 20 female).

All the participants were non-right-handed, according to the Edinburgh Handedness Inventory^{27,28}. There were no significant between-group differences in age ($t_{84} = 1.28$; $P = 0.2$), sex ($\chi^2 = 0.11$; $P = 0.74$), or general intelligence by WAIS-IV matrix reasoning subtest²⁹ ($t_{84} = 1.8$; $P = 0.08$). The participants had no history of any neurological or psychiatric disorders or head injury with loss of consciousness. Written informed consent was obtained from all participants following a protocol approved by Universitat Jaume I. All methods were carried out in accordance with approved guidelines and regulations.

2. Experimental Design

Participants were recruited via multiple advertisements across Castellón and Valencia universities (bulletin boards, mass emailing, etc.) asking for the collaboration of left-handers in an fMRI brain study. Persons older than 36 years or younger than 16 years were discarded. Valid participants were cited for a first fMRI session, in which they completed the verb generation task and the resting-state acquisition. During this session, we used the BrainWave software (GE HealthCare Technologies Inc.) to visualize real-time data and roughly categorize participants as potentially typical or potentially atypical. 43 participants were found potentially atypical for language lateralization, and were invited to a second fMRI session, along with 43 potentially typical participants. During this second session, participants completed the stop-signal task (scanner) and a reading skill test (out of scanner). Also, data regarding schizotypy personality and autistic spectrum was gathered via self-questionnaires. Note that classification of participants as potentially typical or potentially atypical was used for screening purposes, and it did not completely match the final assessment.

2.1 Verb generation Task

Expressive language function was measured via fMRI during a computerized verb generation task^{30,31} that consists of a 2-block design paradigm with intercalating activation and control blocks. In the activation blocks, participants are asked to overtly say the first verb that comes to mind when visually presented with a concrete noun. In the control blocks, they have to read visually-presented letter pairs aloud. This task lasted for 6 minutes, with a block duration of 30 seconds (6 activation blocks and 6 control blocks), a stimulus duration of 1500 ms, and a blank inter-stimulus interval of 1500 ms. Before entering the scanner, participants practiced with a different version of the task for 1 minute. Stimuli were presented using MRI-compatible goggles

(VisuaStim Digital, Resonance Technology Inc.), and responses were recorded with a noise-cancelling microphone (FOMRI III+, Optoacoustics Ltd.) to verify that each participant was engaged correctly in the task.

2.2 Stop-signal Task

Response inhibition was measured via fMRI during a computerized stop-signal task adapted from another study²³. The task consists of an event-related design paradigm with ‘Go’ and ‘Stop’ trials. In the ‘Go’ trials, participants are asked to manually answer by pressing a button if the visually presented string is a word (has a meaning) or a pseudoword (mimics typical word structure but has no meaning). Words require an index button press, whereas pseudowords require a thumb button press. However, on the ‘Stop’ trials, the string is followed by a ‘beep’ noise that signals to the participants to inhibit their response and refrain from pressing any button. The instructions emphasized the importance of going correctly and stopping correctly. However, participants were asked to respond as quickly and accurately as possible and avoid withholding their responses in anticipation of a possible ‘beep’.

This task took 13 minutes and 28 seconds, and it was divided into two runs that lasted 6 minutes and 14 seconds each, separated by a 1-minute rest. Each run consisted of 135 trials, of which 32 (23.7%) were ‘Stop’ trials. All the trials randomly presented words or pseudowords. The trial structure consisted of a fixation crosshair (500 ms), a word or pseudoword (1000 ms, during which the ‘beep’ may or may not be presented at some point), and a blank inter-stimulus interval (ranging from 500 to 4000 ms, sampled from an exponential distribution truncated at 4000 ms, with mean of 1000 ms). The Stop Signal Delay or SSD (the amount of time after the onset of the word or pseudoword when the ‘beep’ is presented during ‘Stop’ trials) changed dynamically during the task after each ‘Stop’ trial, depending on whether inhibition was successful (+25 ms) or unsuccessful (–25 ms), with a minimum of 100 ms and a maximum of 800 ms.

The lower the SSD, the easier it is to inhibit the response, and vice versa. Hence, a dynamic SSD normalizes the task’s difficulty across all participants based on their performance, aiming at a 50% successful inhibition rate. The SSD used for the first ‘Stop’ trial was estimated for each participant based on their practice session before entering the scanner. This practice session lasted 6 minutes and used a different set of words and pseudowords. Stimuli were presented using MRI-compatible goggles and headset (VisuaStim Digital, Resonance Technology Inc.), and responses were recorded via an MRI-compatible response-grip (ResponseGrips, NordicNeuroLab). The ‘beep’ volume was kept at a comfortable level and was constant across participants.

2.3 Resting-state

Functional connectivity was measured via fMRI during a resting-state session³². In this paradigm, participants were presented with a fixation crosshair and instructed to just lie in the scanner with their eyes open and try not to sleep or think about anything in particular. This session lasted for 7 minutes. Seven participants did not complete this session due to time constraints, and they were subsequently removed from the functional connectivity analyses.

2.4 Behavioral measures

Individual inhibition speed was estimated by calculating the Stop-Signal Reaction Time (SSRT)³³. SSRT was computed as the difference between the median reaction time (RT) on correct ‘go’ trials and the mean SSD on the stop-signal task. It should be noted that technical problems involving the MRI-compatible response-grip invalidated the SSD data of four participants, and so their SSRT was estimated using the SSD data from the practice session instead.

Reading skill was evaluated using the word reading subtest of the PROLEC-SE- R³⁴, a battery that assesses reading processes in Spanish. Participants had to read four lists of words consisting, respectively, of short familiar words, long familiar words, short unfamiliar words, and long unfamiliar words. Responses were recorded with a microphone, and their accuracy and speed were subsequently measured by computing length (performance during short words – performance during long words) and familiarity (performance during familiar words – performance during unfamiliar words) indicators.

Schizotypal traits were explored with the SPQ³⁵, a preclinical self-report questionnaire modeled after DSM-III criteria³⁶. This questionnaire evaluates schizotypy based on the three-dimensional model: disorganized traits, cognitive- perceptual traits, and interpersonal traits. For two participants, these data were not collected due to time constraints.

Autistic spectrum traits were explored with the AQ³⁷, a preclinical self-report questionnaire modeled after DSM-IV criteria³⁸. This questionnaire evaluates autistic spectrum based on the “triad” of impaired communication, impaired social skills, and a restricted and repetitive way of acting³⁹, plus two sub-scales for imagination and attention to detail. For two participants, these data were not collected due to time constraints.

3. Image acquisition and processing

Images were acquired on a 3T General Electric Signa Architect magnetic resonance imaging (MRI) scanner. A 3D structural MRI was acquired for each subject using a T1-weighted magnetization-prepared rapid gradient-echo sequence (TR/TE = 8.5/3.3 ms; matrix = 512 × 512 × 384; voxel size = 0.47 × 0.47 × 0.5). For the fMRI, a gradient-echo T2*-weighted echo-planar imaging sequence was used in the acquisition of 150 functional volumes on the verb generation task (TR/TE = 2500/30 ms; matrix = 64 × 64 × 30; voxel size = 3.75 × 3.75 × 4), and a different sequence was used in the acquisition of 374 functional volumes during the stop-signal task and 210 functional volumes in the resting- state (TR/TE = 2000/30 ms; matrix = 64 × 64 × 27; voxel size = 3.75 × 3.75 × 4.5).

Task functional images were processed using the Statistical Parametric Mapping software package (SPM12; Wellcome Trust Centre for Neuroimaging, London, UK). Preprocessing followed the default pipeline and included: (a) alignment of each participant’s fMRI data to the AC-PC plane by using the anatomical image; head motion correction, where the functional images were realigned and resliced to fit the mean functional image; (c) co-registration of the anatomical image to the mean functional image; (d) re-segmentation of the transformed anatomical image using a symmetric tissue probability map; (e) spatial normalization of the functional images to the MNI (Montreal Neurological Institute, Montreal, Canada) space with 3 mm³ resolution; and (f) spatial smoothing (FWHM = 4 mm). The general linear model for the verb generation task was defined for each participant by contrasting *activation > control blocks*.

The general linear model for the stop-signal task was defined for each participant by contrasting *correct ‘stop’ > correct ‘go’ trials*. For both tasks, the BOLD (Blood-Oxygen-Level-Dependent) signal was estimated by convolving the task’s block/trial onsets with the canonical hemodynamic response function. Six motion realignment parameters extracted from head motion preprocessing were included as covariates of no interest, and a high-pass filter (128 s) was applied to the contrast images to eliminate low-frequency components. After that, mean BOLD values were extracted from the stop-signal contrast images for the main components of the inhibitory control network: pars opercularis, pars triangularis, preSMA, and STN. Pars opercularis and pars triangularis of the IFC were defined following the criteria of the Harvard-Oxford atlas^{40–43}. For the PreSMA region, we used the SMA region, as defined by the same atlas, but including only voxels anterior to MNI Y = 0⁴⁴. STN was defined by a 10mm box centered at MNI coordinates 10, -15, -5^{22, 45}. Lastly, we computed whole-brain voxel-wise functional asymmetry maps from the stop-signal

contrast images. To do so, stop-signal contrast images were flipped at midline, inverting the right and left hemispheres, and subsequently subtracted from the original unflipped contrast images⁴⁶.

Resting-state functional images were processed using the Data Processing Assistant for the Resting-State toolbox (DPARSFA)⁴⁷, which is based on SPM and the Data Processing & Analysis of Brain Imaging toolbox (DPABI)⁴⁸. Preprocessing steps included: (a) slice-timing correction for interleaved acquisitions; (b) head motion correction (no participant had a head motion of more than 2 mm maximum displacement in any direction or 2° of any angular motion throughout the scan); (c) co-registration of the anatomical image with the mean functional image; (d) new segmentation to DARTEL; (e) removal of nuisance variance through linear regression: six parameters from the head motion correction, white matter signal, cerebrospinal fluid signal, and global mean signal; (f) spatial normalization to the MNI (3 mm³); (g) spatial smoothing (FWHM = 4 mm); (h) removal of the linear trend in the time series; (i) band-pass temporal filtering (0.01–0.1); (j) normalization to a symmetric template; (k) Voxel-Mirrored Homotopic Computation (VMHC)⁴⁹, calculated as the Pearson correlation coefficient of every voxel with its hemispheric counterpart; and (l) normalization of all voxel-wise time courses to Fisher z values. Finally, mean VMHC values were extracted for the pars opercularis and pars triangularis regions, as defined by the Harvard-Oxford atlas^{40–43}.

Structural images were processed via voxel-based morphometry (VBM) using the CAT12 toolbox⁵⁰, which is based on SPM. Preprocessing steps followed the recommended pipeline and included: (a) segmentation into grey matter, white matter, and cerebrospinal fluid; (b) registration to the ICBM standard template; modulated normalization of grey matter and white matter segments to the MNI template; (d) spatial smoothing (FWHM = 6 mm); and (e) extraction of Regions of Interest (ROIs) values from native space. Three ROIs were delimited by the voxels mapped as genu, body and splenium of the corpus callosum according to the Mori atlas⁵¹. Total intracranial volume was also extracted for use as covariate of no interest.

4. Individual assessment of functional lateralization

Individual functional lateralization was assessed by calculating the Laterality Index (LI) on the unflipped contrast images^{30, 31}. We used the bootstrap method implemented in the LI-toolbox⁵², based on SPM. LI is a proportion of the brain activation between the two hemispheres, thus giving us information about the direction and degree of hemispheric specialization during a particular function in a single individual. LI ranges from +1 (totally leftward function) to -1 (totally rightward function). For both the verb generation task and the stop-signal task, we explored the LI of the inferior frontal region roughly corresponding to the classic Broca's area: pars opercularis and pars triangularis of the inferior frontal gyrus, according to the Harvard-Oxford atlas^{40–43}. This region is critical for both language production³ and response inhibition²⁴, depending on the hemisphere. During the verb generation task, language production was classified as typically lateralized (LI higher than +0.4) or atypically lateralized (LI lower than +0.4). We used +0.4 as a cut-off point (contrary to the more traditional +0.2), based on previous findings that emphasized the importance of lateralization strength when grouping individuals^{6,53}.

5. Statistical analyses

We performed different analyses to investigate the hypothesis that the atypical group would show a mirrored brain organization during the stop-signal task.

First, a series of ROI analyses were employed to compare the functional asymmetry of the main components of the inhibitory control network (during the stop-signal task) between typically and atypically lateralized groups. Hence, four repeated-measures MANOVAs (one for the mean BOLD value inside each ROI: pars opercularis, pars triangularis, preSMA, and STN) were calculated,

including Hemisphere (left/right) as within-subject factor and Group (typical/atypical) as between-subject factor. All p -values were FDR-corrected for four comparisons. This analysis was complemented with whole-brain functional asymmetry contrast maps from the stop-signal task that were compared in typically and atypically lateralized groups based on language production. To do so, we performed a whole-brain two-sample t -test (voxel-wise $P < 0.001$; FWE cluster-corrected at $P < 0.05$) via SPM12. Next, a Pearson's correlation was carried out between the LI from the verb generation task and the LI from the stop-signal task used in the individual lateralization classification ($P < 0.05$, 1-tailed). Finally, we explored the functional overlap between both tasks in the subgroup of participants that presented an ambilateral organization of at least one function ($n = 48$). To do so, we overlaid two one-sample t -tests restricted to the IFC: one for the stop-signal task (voxel-wise $P < 0.001$; FWE cluster-corrected at $P < 0.05$), and one for the verb generation task (voxel-wise $P < 0.05$; uncorrected). Functional overlap was then expressed as the proportion of overlapped voxels for every function and hemisphere.

Using the LIs from the verb generation and stop-signal tasks, we tested if a linear relationship existed between the functional lateralization of an individual, and several behavioral tests and neuroanatomical measures. Partial Spearman's correlations were computed between the LI of both tasks and: stop-signal task performance (RT, accuracy, and SSRT), SPQ derived-scores, AQ derived-scores, reading test performance (length and familiarity effects on speed and accuracy), functional connectivity measures (VMHC), and white matter volume of the callosal genu. Age and general intelligence were included as covariates in all analyses. Total intracranial volume was also included as covariate in the volumetric analysis.

Results

1. Inhibitory control shifts its lateralization according to language production

Inhibitory control components during the stop-signal task were defined by four Regions Of Interest (ROIs) and subsequently analyzed by computing MANOVAs, including Hemisphere as within-subject factor and Group (typical/atypical) as between-subject factor. Results showed that all the tested structures presented a statistically significant Hemisphere $\times$ Group interaction in their BOLD signal during the 'correct stop > correct go' condition (pars opercularis $F_{1,84} = 25.59$, $P < 0.001$; pars triangularis $F_{1,84} = 19.95$, $P < 0.001$; preSMA $F_{1,84} = 7.63$, $P = 0.007$; STN $F_{1,84} = 14.24$, $P < 0.001$; FDR-corrected for 4 comparisons; see **Fig 1a**). In the atypical group, all the regions seemed to have a shift towards leftward lateralization of brain function.

A voxel-wise whole-brain analysis of the functional asymmetry maps during the stop-signal task was performed to complement the ROI results. Two-sample t -test comparisons of the typical and atypical groups during the 'correct stop > correct go' condition showed a differential asymmetry pattern in all the ROI-corresponding areas, while also revealing differences in other cortical and subcortical regions, including the angular gyrus, middle temporal gyrus, and caudate (**Fig 1b**). Specifically, the typical group presented a rightward activity in these regions, whereas the atypical group exhibited a leftward shift (**Fig 1c**).

Finally, the Pearson's correlation between the frontal LIs (pars opercularis + pars triangularis) from the verb generation task and the stop-signal task revealed a strong inverse correlation ($r_{84} = -.58$; one-tailed $P < 0.001$; $R^2 = .34$) (**Fig 2**). Out of 50 typically lateralized for language, 33 (66%) presented a typical right lateralization for inhibitory control, and 17 (34%) showed an atypical (left-ambilateral) lateralization. Among the 36 individuals from the atypical group, only 8 (22.2%) presented a typical lateralization during the stop-signal task, whereas 28 (77.8%) showed an atypical lateralization. Importantly, no participant presented a leftward + leftward or rightward + rightward segregation pattern for language production and response inhibition. Strong

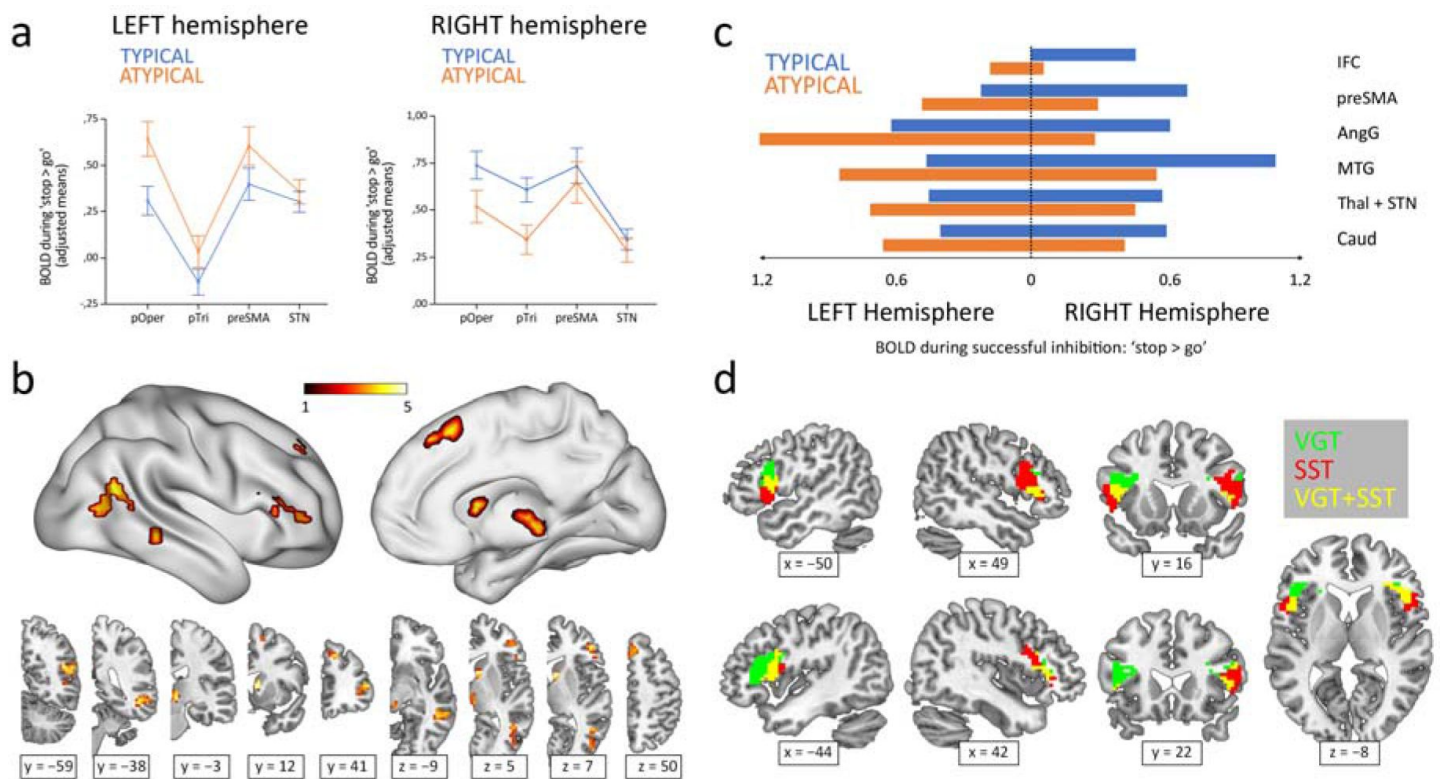


Fig 1.

Hemispheric lateralization of inhibitory control according to language lateralization. **(A)** ROI analysis of the main components of the inhibitory control network. Graphs depict adjusted mean BOLD signal during successful inhibitions on the stop-signal task ('stop > go' contrast) for both hemispheres and both groups. All four structures showed significant Hemisphere $\times$ Group interactions. Error bars represent one standard error. pOper = pars opercularis, pTri = pars triangularis, preSMA = presupplementary motor area, and STN = subthalamic nucleus. **(B)** Voxel-wise whole-brain analysis of functional asymmetry. Significance maps (voxel-wise $P < 0.001$; FWE cluster-corrected at $P < 0.05$; color bar represents t value) are displayed in three-dimensional reconstructions plus coronal and transversal slices using MNI space. **(C)** Mean BOLD values of the significant regions found in the voxel-wise whole-brain analysis. Graphic depicts BOLD values for every region, hemisphere, and group. IFC = inferior frontal cortex, preSMA = presupplementary motor area, AngG = angular gyrus, MTG = middle temporal gyrus, Thal = thalamus, STN = subthalamic nucleus, Caud = caudate. **(D)** Functional overlap between language production and inhibitory control in the IFC of ambilateral participants. Conjunction maps for inhibition (voxel-wise $P < 0.001$; FWE cluster-corrected at $P < 0.05$) and language (voxel-wise $P < 0.05$; uncorrected) are displayed in coronal, sagittal and transversal slices using MNI space. VGT = verb generation task, SST = stop-signal task.

segregation of both functions into different hemispheres was observed in 38 participants (44.2%). The remaining 48 individuals (55.8%) presented an ambilateral representation of one or both functions, with an important level of spatial overlap in their activations (see [Fig 1d](#)). In this ambilateral group, language production presented an overlap in 32.3% of its voxels in the left hemisphere, and in 58.6% of its voxels in the right hemisphere. Inhibitory control showed an overlap in 57.2% of its voxels in the left hemisphere, and in 22.8% of its voxels of the right hemisphere.

2. Laterality Indexes correlate to interhemispheric connectivity and preclinical markers

Correlational analyses revealed certain links between the functional lateralization of both tasks and behavioral, neuroanatomical and connectivity measures ([Table 1](#)).

First, we studied how the structural and functional connectivity between both IFC behaved in relation to LI. Volume in the genu, body and splenium of the corpus callosum was inversely related with language LI (respectively: $\rho_{83} = -.25$, $P = 0.02$; $\rho_{83} = -.26$, $P = 0.02$; and $\rho_{83} = -.25$, $P = 0.02$). However, only the callosal genu was found related with inhibition LI ($\rho_{83} = .25$, $P = 0.02$). That is, the volume of the callosal genu increased as functional organization of the IFC became more atypical, extending this effect to the whole corpus callosum when considering exclusively language lateralization. On the same line, ROI interhemispheric functional connectivity analyses at rest revealed that VMHC of the pars triangularis also increased as a function of inhibition LI ($\rho_{77} = .26$, $P = 0.02$), but a similar relation failed to be found for language LI ($\rho_{77} = -.12$, $P = 0.3$). No statistically significant linear relationships were found when exploring the VMHC of the pars opercularis.

Behaviorally, performance (RT, SSRT and accuracy) during the scanner stop- signal task did not correlate with language and inhibition LIs. Regarding the reading test, accuracy for word length and word familiarity were found correlated with both LIs (length and language: $\rho_{84} = -.24$, $P = 0.03$; length and inhibition: $\rho_{84} = .21$, $P = 0.04$; familiarity and language: $\rho_{84} = -.23$, $P = 0.03$; familiarity and inhibition: $\rho_{84} = .3$, $P = 0.01$). That is, an atypical organization was related with a higher rate of errors when reading long words and unfamiliar words. No significant correlations were found when exploring reading speed.

SPQ score was found significantly related with inhibition LI ($\rho_{82} = .25$, $P = 0.02$), but it did not reach statistical significance when paired with language LI ($\rho_{84} = -.19$, $P = 0.08$). AQ, on the other hand, presented an association with language LI ($\rho_{84} = -.25$, $P = 0.02$) but not with inhibition LI ($\rho_{84} = .08$, $P = 0.5$).

Discussion

The present study employed the stop-signal task to investigate brain areas involved in the inhibitory control function in two different groups of individuals: typically and atypically lateralized for language production. As expected, the atypical participants showed, as a group, a mirrored brain organization of the inhibitory control function compared to the typical group. This leftward organization affected the entire inhibitory control network, including the IFC, preSMA, and STN. However, some participants manifested a clear overlap in the control of language and inhibitory functions. Our results also demonstrate that atypical organization of language production is associated with an increased white matter volume of the corpus callosum, and that atypical lateralization of inhibitory control is related with a higher interhemispheric functional

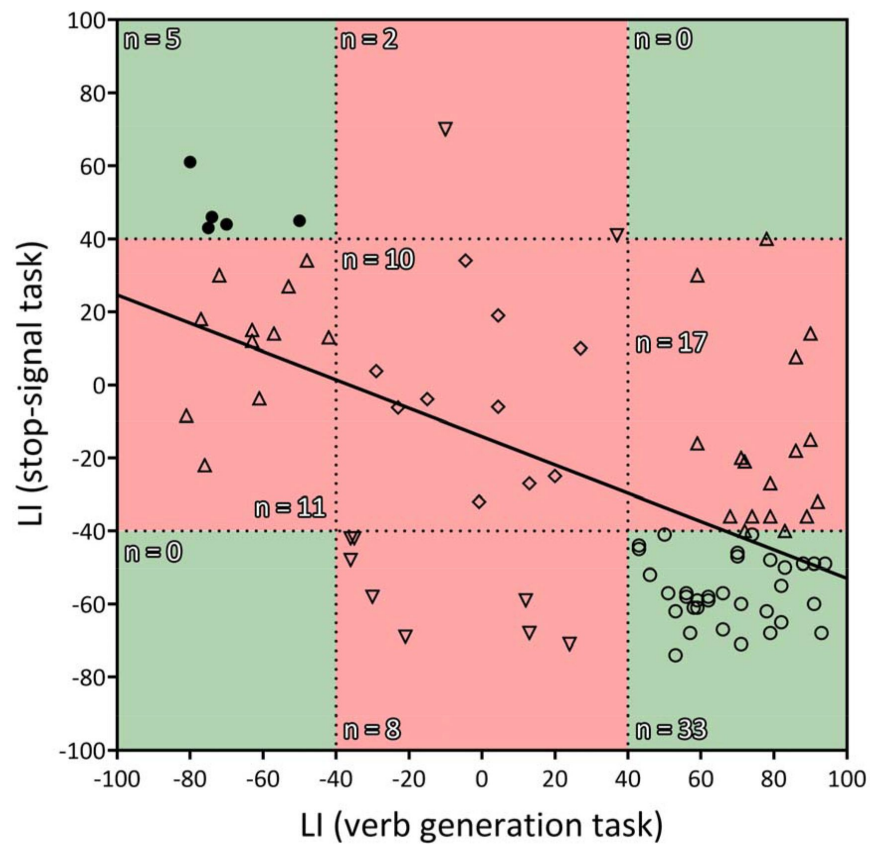


Fig 2.

Correlation between the LIs of the verb generation and stop-signal tasks. $r = -.583$, one-tailed $P < 0.001$, $R^2 = .339$. Negative values indicate rightward lateralization, whereas positive values indicate leftward lateralization. Segregated and ambilateral phenotypes are also depicted according to the background color. The green area corresponds to segregated individuals (both functions strongly lateralized), and the red area corresponds to integrated individuals (at least one function ambilaterally controlled). Numbers inside each quadrant denote the number of individuals (n) contained in it. Each individual datapoint is symbolized according to its functional organization: $\circ$ = typical segregation; $\bullet$ = reversed segregation; $\triangle$ = ambilateral inhibition; ∇ = ambilateral language; $\diamond$ = ambilateral language and inhibition.

	Verb generation task LI		Stop-signal task LI	
	Spearman's ρ	P	Spearman's ρ	P
Callosal genu volume	-.249	.021*	.251	.02*
Callosal body volume	-.258	.017*	.132	.224
Callosal splenium volume	-.249	.021*	.118	.279
Pars Opercularis VMHC	-.062	.584	.148	.192
Pars Triangularis VMHC	-.119	.295	.258	.022*
'Go' reaction time	.125	.251	-.041	.71
'Go' accuracy	-.076	.489	.039	.724
SSRT	.173	.111	-.038	.726
Reading length accuracy	-.237	.028*	.214	.047*
Reading familiarity accuracy	-.232	.032*	.301	.005**
Reading length time	.054	.622	-.184	.068
Reading familiarity time	.091	.403	-.198	.091
SPQ	-.194	.078	.247	.023*
AQ	-.247	.023*	.075	.498

Table 1.

Spearman's partial correlations between task LIs and neuroanatomical plus behavioral variables. General intelligence and age were included as covariates of no interest. Callosal volume correlations were additionally corrected for total intracranial volume.

coupling of the IFC. Behaviorally, atypical lateralization was not associated with performance on the task, but an association existed with worse reading accuracy, and more schizotypy and autistic traits.

Our first relevant result is that the group of individuals with atypical lateralization of language presented a mirrored brain organization during the stop-signal task compared to the typically lateralized group. Remarkably, the hemispheric asymmetry differences obtained affected the entire inhibitory control brain network²⁵. The inferior frontal cortex, or IFC, associated with the initiation of the stop command and the main cortical area of this system, showed a clear lateralization effect, with the typical group almost exclusively using the right part and the atypical group mostly activating the left part. The presupplementary motor area, or pre-SMA, related to the implementation of the stop command, was more bilaterally involved in both groups, although atypicals activated the left part more and typicals activated the right part more. Importantly, the hemispheric reversal of organization also affected the subcortical structures participating in this network, namely the STN and thalamus. These structures displayed a lateralization pattern similar to the preSMA, with a more pronounced bilateral engagement but higher involvement of the right part in typicals and of the left part in atypicals. Thus, this is the first demonstration that an entire cerebral network controlling a cognitive function shows a completely flipped organization in an atypical population for language, which supports the Causal hypothesis of lateralization^{10, 11}. Moreover, this result reveals that hemispheric lateralization goes beyond cerebral cortices, and even interhemispheric connectivity, through the corpus callosum, given that it also involves frontal-subcortical circuits. Additionally, the specificity of this result to the areas of the proposed network is another element that supports the relevance of this system in inhibitory control.

It should be noted that this pattern of atypical brain specialization was observed in the absence of a relation between functional lateralization and performance differences during the ‘go’ or ‘stop’ conditions. Even though the ‘go’ condition involves a lexical decision requiring language processing, and the ‘stop’ condition consists of inhibiting a linguistic process, the functional organization of the IFC during the tasks did not affect the response speed (RT and SSRT) or the ‘go’ accuracy in either case. Thus, brain organization for language and inhibition, as initially defined in our study, and cognitive efficiency are not directly related, based on our data. This finding is consistent with previous reports showing that cognitive deficits appear only in specific cognitive domains^{17, 54}.

Importantly, despite observing a complimentary relationship between language production and inhibitory control at the group level, not all the individuals presented a strong and segregated distribution of these two functions. The vast majority of hemispherically segregated individuals showed the typical segregation of language in the left hemisphere and inhibitory control in the right hemisphere, whereas only a low percentage displayed the opposite segregated organization. The rest, in agreement with similar reports on other functions¹⁹, lacked a clear hemispheric lateralization of language production, inhibitory control, or both. In all cases, this implied some level of overlapping or sharing of the same area of the inferior frontal cortex. Both the low ratio of atypical segregation and the high ratio of bilateral inhibitory control could be explained by the fact that typical right hemispheric functions (such as inhibition) seem to be more bilaterally represented than typical left functions (such as language)⁵⁵. Some conclusions can be extracted from this: (1) Although infrequent, a strong atypical segregation of the two functions is possible; (2) Our data have not revealed any case with a segregation of both frontal cognitive functions in the same hemisphere; and (3) A large percentage of participants with a strong left lateralization of language presented a bilateral control of inhibition, showing that a typical dominance for language does not imply a strongly lateralized organization of other cognitive functions. The existence of exceptions to the group pattern of mirrored brain organization requires us to use caution when interpreting these data as supporting the causal hypothesis. Although the negative correlation between the lateralization indexes during the verb generation task and the stop-signal task is strong at the group level, some individual data support the statistical hypothesis¹².

instead. Given that we found no cases of both cognitive functions completely sharing a hemisphere (see **Fig 2**), we propose that causal-supporting vs. statistical-supporting results reflects two different pathways to cognitive control. On the one hand, the segregated pathway (in line with the Causal hypothesis) develops due to the cerebral bias towards lateralizing certain cognitive functions in a single hemisphere. On the other hand, the integrated pathway (in line with the Statistical hypothesis), appears when ontogenetic development towards strong lateralized control fails in some way.

The corpus callosum is the main cerebral structure in interhemispheric connectivity, with the genu area being responsible for the connectivity between the two inferior frontal gyri⁵⁶. Phylogenetic data have demonstrated a negative association between the corpus callosum volume and hemispheric specialization, suggesting that the ontogenetic development of the brain is supported by a decrease in interhemispheric connectivity to potentiate and establish intrahemispheric connectivity and hemispheric specialization⁵⁷. There is evidence of this mechanism in language development⁵⁸ and in agenesis of the corpus callosum, which has been associated with lower lateralization of language and worse language performance^{59, 60}. In line with this proposal, we have replicated the relation between callosal enlargement and atypical functional organization found in a previous study⁵³. Moreover, in our data, this structural correlation was accompanied by an association between atypical lateralization of inhibitory control and strength of interhemispheric functional connectivity in the pars triangularis of the IFC.

As hypothesized, when considering the functional lateralization during both tasks, atypical organizations were found linearly related with higher scores on a schizotypy test (only for inhibitory control), higher scores on an autistic spectrum test (only for language production), and a higher rate of reading errors (both language production and inhibitory control). The first result is consistent with previous structural, functional, and behavioral data that have shown reduced language lateralization in schizophrenia patients^{8, 61}. In the case of schizotypy, the evidence for reduced lateralization of cognitive functions is mixed and may respond to differences in the methodology employed⁶².

However, consistently with the present results, this condition has been previously associated to impaired behavioral and neural processing during the stop-signal task⁶³, and to an abnormal frontal functional asymmetry⁶⁴. The second result relates atypical language lateralization with the presence of autistic traits, a result consistent with data obtained in autism⁶⁵. The third result shows that atypical organization was related with a higher rate of reading errors when considering word length and word familiarity. The length effect has been considered a pathognomonic symptom of reading disorders such as developmental dyslexia⁶⁶ and pure alexia⁶⁷. These disorders have also been characterized by a weak language lateralization⁷. Thus, presented results support the hypothesis that atypical hemispheric specialization is related with worse cognitive performance in the linguistic domain, and even preclinical traits of some neurodevelopmental disorders among healthy population.

Data availability

All data presented in this study are included in the following public dataset:
<https://doi.org/10.6084/m9.figshare.22047299>.

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Author contributions

Conceptualization, C.A. and M.-A.P.; Methodology, E.V.-R. and C.A.; Formal Analysis, E.V.-R.; Investigation, E.V.-R., C.C.-M., and L.M.-M.; Data Curation, E.V.-R. and C.C.-M.; Writing, E.V.-R., C.A. and M.-A.P.; Supervision, C.A.; Project Administration, C.A.

Declaration of interests

The authors declare no competing interests.

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Reviewer #1 (Public Review):

The authors' aim was to test to what extent atypical organization of language is associated with a mirrored brain organization of other cognitive functions. In particular, they focused on the inferior frontal gyri (IFG) by studying the inhibitory control network. This allowed them to directly test the support for the Causal hypothesis of hemispheric specialization, arguing for fast sequences of cognitive processes being better performed by a single hemisphere, versus the Statistical hypothesis of lateralization, postulating an independent lateralization of each cognitive function.

Previous studies on this topic did not focus on functions involving homotopic language regions. This limitation is bypassed in this study by assessing inhibition with a Stop-Signal Task which also engages the IFG in the contralateral site to the verb generation task. By studying a combination of structural and functional information, in addition to the activation contrasts, the authors are able to test whether atypical organization is accompanied by stronger interhemispheric connectivity. Although relying mainly on correlations and lacking

important methodological information that may be critical to understand the reported effects, the results are quite straightforward. However the bilingual/monolingual status and gender of the participants is not reported which might affect the relationship between language and inhibitory control.

The conclusions of the paper are supported by the data. With their design, the authors observed that, as a group, individuals with atypical organization show a mirror organization of the whole inhibitory network to the contralateral site, supporting the Causal hypothesis at the group level. However, individual data support the Statistical hypothesis, since the segregation between language and inhibition was not observed in all individuals and a variety of configurations in bilateral and bilateral organisation of language and inhibition were also observed.

The results of this study have important implications for our understanding of the independence between different cognitive functions which is crucial when addressing brain damage and rehabilitation. This aspect also indirectly speaks to researchers interested in evolution and in bilingualism and its relation to cognitive control. These aspects are not discussed but incorporating them would broaden the interest of the paper beyond the current implications mentioned.

Reviewer #2 (Public Review):

Language skills are traditionally associated with a network of brain regions in the left hemisphere. In this intriguing study, Esteban Villar-Rodríguez and collaborators examined if atypical hemispheric lateralization for language determines the functional and structural organisation of the network for inhibitory control as well as its relationship with schizotypy and autistic spectrum traits. The results suggest that individuals who have atypical lateralisation of the language function have also an atypical (mirrored) lateralisation of the inhibitory control network, compared to the typical group (individuals with left-lateralised language function). Furthermore, the atypical organization of language production is associated with a greater white matter volume of the corpus callosum, and atypical lateralization of inhibitory control is related to a higher interhemispheric functional coupling of the IFC, suggesting a link between atypical functional lateralisation (language and inhibitory control) and structural and functional changes in the brain.

This study also provides interesting evidence on how atypical language lateralisation impacts some aspects of language behaviour (reading), i.e., atypical lateralization predicts worse reading accuracy. Furthermore, the results suggest an association between atypical lateralization and increased schizotypy and autistic traits.

The strength of this work is that it presents a collection of measurements on the same individuals (including task-related behavioural, functional and structural neuroimaging measures) to reveal if (and how) atypical language lateralisation might be associated with: (1) atypical neural organisation of other non-linguistic cognitive systems, (2) behavioural performance associated with language tasks, and finally (3) personality traits. As such the results presented in this manuscript have the potential to be informative for various disciplines. For instance, if clarifications/corrections are provided (see below), the results might provide some insight into the role of the right hemisphere for language processing in healthy individuals as well as patient populations with acquired linguistic impairment including stroke and dementia.

One important weakness of this manuscript is that several areas, including the characteristics of participants tested, and the hypotheses/predictions, are underspecified or incomplete. Furthermore, in some cases the types of analysis do not seem to be appropriate for addressing the questions of the present study and very little explanation for those choices is provided.